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Linear Discriminant Analysis

The background of the slide is a dark blue field filled with a complex network of glowing blue lines and dots, resembling a circuit board or neural network. In the center, a glowing white brain icon is superimposed on a dark square. To the right of the brain, a human hand is visible, with the index finger pointing towards the brain icon.

When to Use LDA

Ideal Scenarios for Applying Linear Discriminant Analysis

Clear Class Boundaries

Best when classes are clearly and linearly separable

Small-to-Moderate Feature Sets

Suitable for datasets with a moderate number of predictors

Continuous Data

Works well with continuous variables, not categorical

Multi-Class Problems

Effective for problems with more than two classes

Required Assumptions for LDA

Conditions That Must Be Met for Accurate Results

Exclusive Classes

- 1 Each data point must belong to only one, non-overlapping class

Normal Distribution

- 2 Features should be normally distributed within each class

Equal Covariance

- 3 All classes must share the same variance-covariance structure

Independent Data

- 4 Observations must be independent of one another

LDA in Practice: Dimensionality Reduction

Simplifying High-Dimensional Data for Better Insights

LDA is a classification and dimensionality reduction tool, ideal for high-dimensional data

- Reduces complex data while keeping class information intact
- Projects data from n features to $(c-1)$ dimensions (c = number of classes)
- Maintains a balance between accuracy and efficiency
- Enables clear 2D or 3D visualisation of data

Use Case: Medical Diagnosis

LDA for Accurate and Explainable Health Insights



Tumour Classification

- Classifies tumours as benign or malignant using MRI features
- Extract quantitative features from medical scans
- Identify key patterns separating harmful and harmless growths



Clinical Decision Support

- Use LDA-powered tools to improve diagnostic accuracy and reduce interpretation time.
- Highlight suspicious regions and provide probability scores as a second opinion.

LDA's transparency helps gain clinical trust and meet regulatory needs

Use Case: Finance & Credit Scoring

LDA for Smarter Lending and Risk Management

Risk Assessment

Banks use LDA to separate high-risk and low-risk borrowers based on income, debt, payment history, and employment

Default Prediction

Improves accuracy of default predictions by 15–20% over traditional methods, reducing financial losses

Regulatory Compliance

LDA's transparency helps explain credit decisions and meet fair lending regulations

Use Case: Bioinformatics

LDA for Analysing Complex Gene Expression Data

LDA helps analyse microarray data with thousands of genes across tissue samples
Identifies gene patterns to distinguish:

- Healthy vs. diseased tissues
- Cancer subtypes
- Treatment responders vs. non-responders

Ideal for biological data with more genes than samples – dimensionality reduction is key

"LDA has become an indispensable tool in our cancer research, helping us identify the genetic signatures that predict treatment response with over 85% accuracy."

— Dr. Sarah Chen, Cancer Research Institute

Additional Real-World Applications

How LDA Powers Multiple Industries



Marketing

- Use LDA to segment customers by behaviour and data
- Discover distinct target groups
- Drive focused marketing efforts
- Raise conversion rates up to 30%



Face Recognition

- Reduce facial data dimensions.
- Maximise class separation.
- Enable fast recognition.
- Support security and photo apps



Quality Control

- Analyse sensor data on production lines
- Detect subtle defects
- Distinguish faulty products
- Improve quality control accuracy



Speech Recognition

- Simplify audio features to key dimensions
- Enhance phoneme distinction in speech.
- Improve recognition accuracy
- Work well in noise

Strengths of LDA

Efficient, Reliable and Easy to Interpret

Computational Efficiency

Enable quick training and classification on basic hardware, making LDA more efficient than complex models like neural networks or SVMs

Dataset Compatibility

Performs exceptionally well on small to medium datasets, where advanced models often overfit. Ideal for research or applications with limited data

Interpretable Results

Generates linear decision boundaries and clear coefficients that show feature importance—making results easy to explain to stakeholders

Accuracy Considerations

Performance Factors

LDA accuracy depends on key factors:

- **Dataset Size:** Accuracy improves with more training samples per class
- **Class Separation:** Less effective when classes overlap significantly
- **Outliers Presence:** Sensitive to extreme values that can shift decision boundaries
- **Class Balance:** Performs best when class sizes are similar

Achieves 70–90% accuracy on well-structured problems where assumptions are met—comparable to more complex models

Limitation: Data Reduction Loss

Trade-offs in Simplifying High-Dimensional Data

Information Sacrifice

- 1 Reduce dimensions from n to $c-1$, which may lose key information in non-linear feature relationships

Projection Limitations

- 2 Miss important class differences if they don't align with between-class variance

Complex Dataset Challenges

- 3 Oversimplify complex feature interactions, leading to lower classification performance

Limitation: Outliers and Variance

Sensitivity to Data Irregularities and Uneven Spread

Vulnerability to Extreme Values

LDA is sensitive to outliers as it depends on means and covariance

- A single extreme value can shift class means noticeably
- Outliers distort covariance estimates, leading to poor projections
- Classification boundaries may be skewed by unusual data points

Homogeneity Requirement

- Assumes equal covariance across classes (homoscedasticity)
- Often not true in real-world data, especially with uneven class sizes or very different class types

Limitation: Assumptions on Data Structure

Challenges When Real-World Data Doesn't Fit LDA's Assumptions



Normality Violation

- Real data often isn't normally distributed, affecting LDA performance
- Even with fixes like Box-Cox, accuracy may still drop



Covariance Equality

- LDA assumes equal covariance, which is often unrealistic
- QDA handles unequal variances better but is more complex



Categorical Variable Issues

- LDA is built for continuous features
- Dummy encoding can disrupt normality and distort covariance

Limitation: Small Datasets and Overfitting

Sample Size Requirements

For reliable LDA implementation, statisticians recommend:

- Minimum 20 samples per group for accurate means
- At least 5× more samples than features to avoid covariance issues
- Balanced class sizes to reduce bias

If these needs aren't met, LDA may perform well in training but fail on new data—showing overfitting



Warning: Perfect training results but poor validation may mean overfitting from too few samples

Best Practices for Using LDA

Steps to Ensure Reliable and Accurate Results



Verify Assumptions

Check for normality and equal covariance; use transformations or other methods if needed



Preprocess Data

Remove outliers, standardise features, and consider PCA to enhance performance



Balance Classes

Apply oversampling or undersampling to handle class imbalances



Compare Models

Evaluate LDA alongside models like QDA, logistic regression, or non-linear methods to choose the best fit

When used with suitable data, LDA is a powerful and interpretable tool widely applied in fields like medicine, finance, and manufacturing

LDA Implementation

Practice Exercise

Problem Statement

Title: Handwritten Digit Classification using LDA on MNIST Dataset

Objective

Build a classification pipeline using LDA for dimensionality reduction, followed by a classifier like Logistic Regression or SVM to predict digits (0–9).

Problem Context

The MNIST dataset consists of thousands of grayscale images of handwritten digits, each represented as a 28×28 pixel matrix (flattened to 784 features). While models like neural networks can learn directly from the high-dimensional space, this project explores the classical statistical approach using LDA to reduce the dimensionality to a maximum of 9 discriminant components (as MNIST has 10 classes), with the goal of improving interpretability and computational efficiency without severely compromising accuracy.

Problem Statement

Goals:

- Apply LDA to reduce MNIST dimensions
- Visualise digit separation in lower dimensions
- Assess classification accuracy post-reduction
- Compare performance with and without LDA

Feature Name	Description
pixel_0 to pixel_783	Grayscale pixel values ranging from 0 to 255, representing the 28x28 image
label	Digit class label (0 to 9)

Q&A Session



Open floor for questions and clarifications.